

FINDINGS.MD

HVAE v1: Emergent Representation Learning on Genomic Data

Status: COMPLETED – Negative Result

This experiment ran to completion. The model trained for 50 epochs.`

It did not produce emergent representations. This document explains why and what was learned.

Research Question

Can a hierarchical variational autoencoder develop self-organised latent structure when trained on genomic sequences without semantic labels or human-imposed categories?

Architecture

- 3-level Hierarchical VAE: 256d → 512d → 1024d latent spaces
- Input: 4096d (1024bp one-hot encoded DNA)
- 28,471,808 parameters
- β -annealing schedule (linear warmup over 20 epochs)
- Optimiser: AdamW, $\text{lr}=1\text{e-}3$
- Data: C. elegans genome (~100Mbp, ~100k windows)

Results

Training Convergence

- Final training loss: 748.1270
- Final validation loss: 748.0401
- Loss flatlined; no meaningful convergence achieved

Latent Space Collapse

- KL divergence (total): 0.0029
- Per-level KL: all ≈ 0
- **Interpretation:** The model learned to bypass the latent distribution.

The reparameterization trick sampled noise that the decoder ignored.

Reconstruction

- Accuracy: 0.00%
- The model did not learn to reconstruct DNA sequences.
- Encoder-decoder weights did not encode meaningful inverse operations.

Generation

- 200 sequences sampled from prior $N(0,1)$
- Mean GC content: 0.00% (target: 36%)
- Generated sequences are statistically indistinguishable from random noise

"Clustering"

- Silhouette scores: ~ 0.001 (level 1), -0.003 (level 3)
- No self-organised clusters emerged
- UMAP projections show no manifold structure

What Went Wrong

1. **Posterior collapse.** β -annealing was insufficient. The KL term was too weak to force structured latents against the high reconstruction pressure of genomic data.

2. **Discrete data in a continuous space.** DNA is categorical (A/C/G/T). One-hot encoding doesn't create a smooth manifold. VAEs assume continuous structure that doesn't exist in genomic sequences.
3. **No inductive bias for sequence.** The architecture used fully-connected layers throughout. No convolutions, no recurrence, no attention – nothing that knows DNA has position-dependent motifs and long-range correlations.

Scientific Value

This negative result directly motivated the SimAE successor architecture.

The key insight: KL divergence *prevents* structure discovery on this data type. A contrastive objective (NT-Xent) that pushes similar sequences together in latent space is a better inductive bias than KL divergence toward $N(0,1)$.

Files

- `src/` – Empty. The aspirational modular structure was never populated.
- `HVAE-TRAINING-FIRST_RUN/Hierarchical_VAE.py` – Original monolithic model
- `scripts/` – CLI training/evaluation/generation scripts (reference `src/` modules that don't exist; non-functional without manual path fixes)
- `notebooks/` – Multiple versions of the same notebook (`_complete`, `_unified`, ``colab``, ``kaggle``, ``v1c``, ``v1g``, ``v1o``, ``v1p``).
- `notebooks/hvae_v1c/` contains the actual training outputs. ``
- `HVAE-TRAINING-FIRST_RUN/` Training outputs: loss curves, UMAP plots, intrinsic dimensionality analysis, screenshots

Conclusion

training dynamics is insufficient to produce self-organised concepts from raw genomic sequence. Structure requires either:`

- Stronger inductive biases (convolutions, transformers)`
- Different objectives (contrastive learning)`
- Different data modalities (language, images with smooth manifolds)`

This experiment is **complete**. No further training will improve the result. The architecture has been superseded by SimAE v1 (contrastive autoencoder, 128d latent, NT-Xent loss).`

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